CT · WEEK 4

CT week 4 patterns: Nested loops, binning and tracing

Nested iterations: comparing pairs · Binning · Tracing: 'What does A represent?'. Save this PDF for revision.

NESTED ITERATIONS: COMPARING PAIRS · IN EVERY PAPER

Every pair exactly once

```
count = 0
while(Table 1 has more rows){
    Read the first row X in Table 1
    Move X to Table 2
    while(Table 1 has more rows){
        Read the first row Y in Table 1
        if(X.CityTown == Y.CityTown){
            count = count + 1
        }
        Move Y to Table 3
    }
    Move all rows from Table 3 to Table 1
}
```

count = number of pairs of students from the same city.

NESTED ITERATIONS: COMPARING PAIRS · IN EVERY PAPER

Nested iterations: comparing pairs

Unordered pairs (each once)

$$(n(n-1))/2$$

n = 30 ⇒ 435

Ordered pairs, X ≠ Y

$$n(n-1)$$

Every X with every Y, itself included

$$n^2$$

Pairs within groups of sizes k_1, k_2, …

$$\Sigma\,(k_i(k_i-1))/2$$

Pairs across two groups

$$k_1 \times k_2$$

NESTED ITERATIONS: COMPARING PAIRS · IN EVERY PAPER

Key points

- Moving X out before the inner loop means X never meets itself and each pair is met once.
- If the inner loop also sees X and the cards before it, each pair is counted twice. Halve it.
- Birthday paradox: with just 23 people there's about a 50% chance two share a birthday (about 70% with 30), because the number of pairs grows like n^2.

NESTED ITERATIONS: COMPARING PAIRS · IN EVERY PAPER

Common traps

⚠ Read whether the question asks for matching pairs or for comparisons made. Comparisons = n(n – 1)/2, whatever the if says.

⚠ 'Either same gender or same city, but not both' is exclusive or: (A and not B) or (B and not A).

BINNING · SOMETIMES ASKED

Key points

- Binning: first sort the cards into bins by a field (city, birth month, gender), then compare pairs only inside each bin.
- Comparisons drop from n(n – 1)/2 to the sum of k(k – 1)/2 over the bins, plus one pass to fill the bins.

BINNING · SOMETIMES ASKED

Binning

Comparisons with bins

$$\Sigma\ (k_i(k_i - 1))/2$$

TRACING: 'WHAT DOES A REPRESENT?' · IN EVERY PAPER

Five-step checklist

- Start value: 0 → count or sum; large → min; True/False → flag.
- Update: + 1 → count; + X.field → sum; = X.field → max, min or 'last seen'.
- Filter: list every condition; note and/or and strict (>) vs non-strict (≥).
- Scope: reset inside the outer loop? Then it's per-card, and the final value belongs to the last card.
- Dry-run 3–4 cards in a trace table (one column per variable) when unsure.

TRACING: 'WHAT DOES A REPRESENT?' · IN EVERY PAPER

Flag pattern

```
found = False
while(Table 1 has more rows){
    Read the first row X in Table 1
    if(X.Total > 250){
        found = True
    }
    Move X to Table 2
}
```

found = True if at least one student scored above 250.

TRACING: 'WHAT DOES A REPRESENT?' · IN EVERY PAPER

Dataset-specific patterns

- Words: a word ending in a full stop closes a sentence. Per-sentence counters reset there.
- Shopping Bills, 'bought both Bread and Milk' (PYQ Jan 2024): set two flags while scanning a bill's items, then count bills where both are True. A ratio like frac is that count ÷ whatever count the question puts in the denominator.

TRACING: 'WHAT DOES A REPRESENT?' · IN EVERY PAPER

Common traps

⚠ Options often differ only by > vs ≥, and vs or, or first vs last. Check those three first.

⚠ A count or sum with no filter is about all cards, not the group the option names.

Notes for every week of CT, free.

Full notes, the pattern sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/ct →

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